# WELCOME

We're Glad You Joined Us Today!

## SOFTWARE ENGINEERING

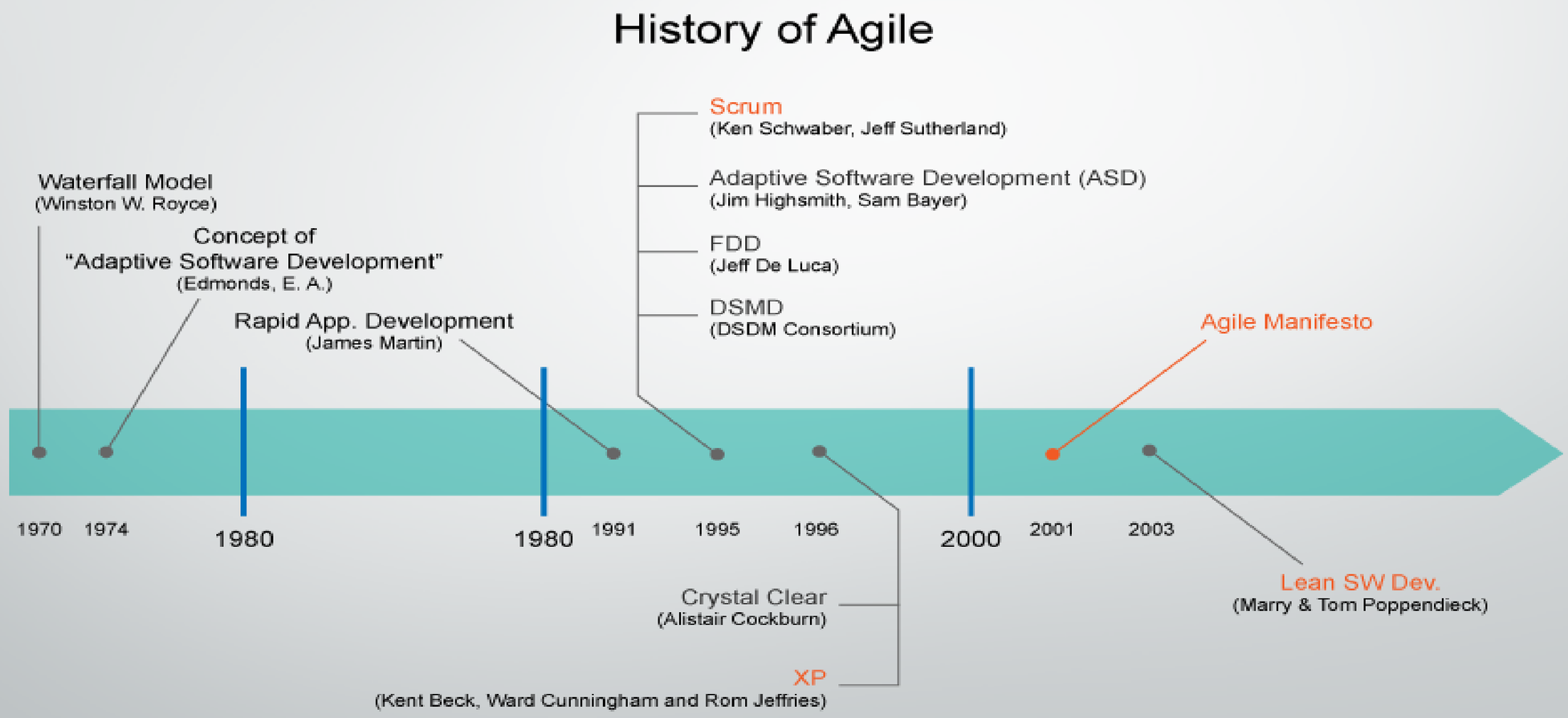
# AGILE SOFTWARE DEVELOPMENT

| SL NO. | CONTENTS |
|---|---|
| 01. | INTRODUCTION TO AGILE DEVELOPMENT |
| 02. | HISTORY OF AGILE |
| 03. | AGILE SOFTWARE DEVELOPMENT |
| 04. | STEPS OF AGILE SOFTWARE DEVELOPMENT |
| 05. | AGILE METHOD EXAMPLES\TYPES OF AGILE FRAMEWORKS |
| 06. | ADVANTAGES OF AGILE SOFTWARE DEVELOPMENT |
| 07. | DISADVANTAGES OF AGILE SOFTWARE DEVELOPMENT |
| 08. | WHEN TO USE THE AGILE MODEL? |

# INTRODUCTION TO AGILE DEVELOPMENT

- Agile Software Development is a Software Development Methodology that values flexibility, collaboration, and customer satisfaction. It is based on the Agile Manifesto, a set of principles for software development that prioritize individuals and interactions, working software, customer collaboration, and responding to change.
- Agile Software Development is an iterative and incremental approach to Software Development that emphasizes the importance of delivering a working product quickly and frequently. It involves close collaboration between the development team and the customer to ensure that the product meets their needs and expectations.

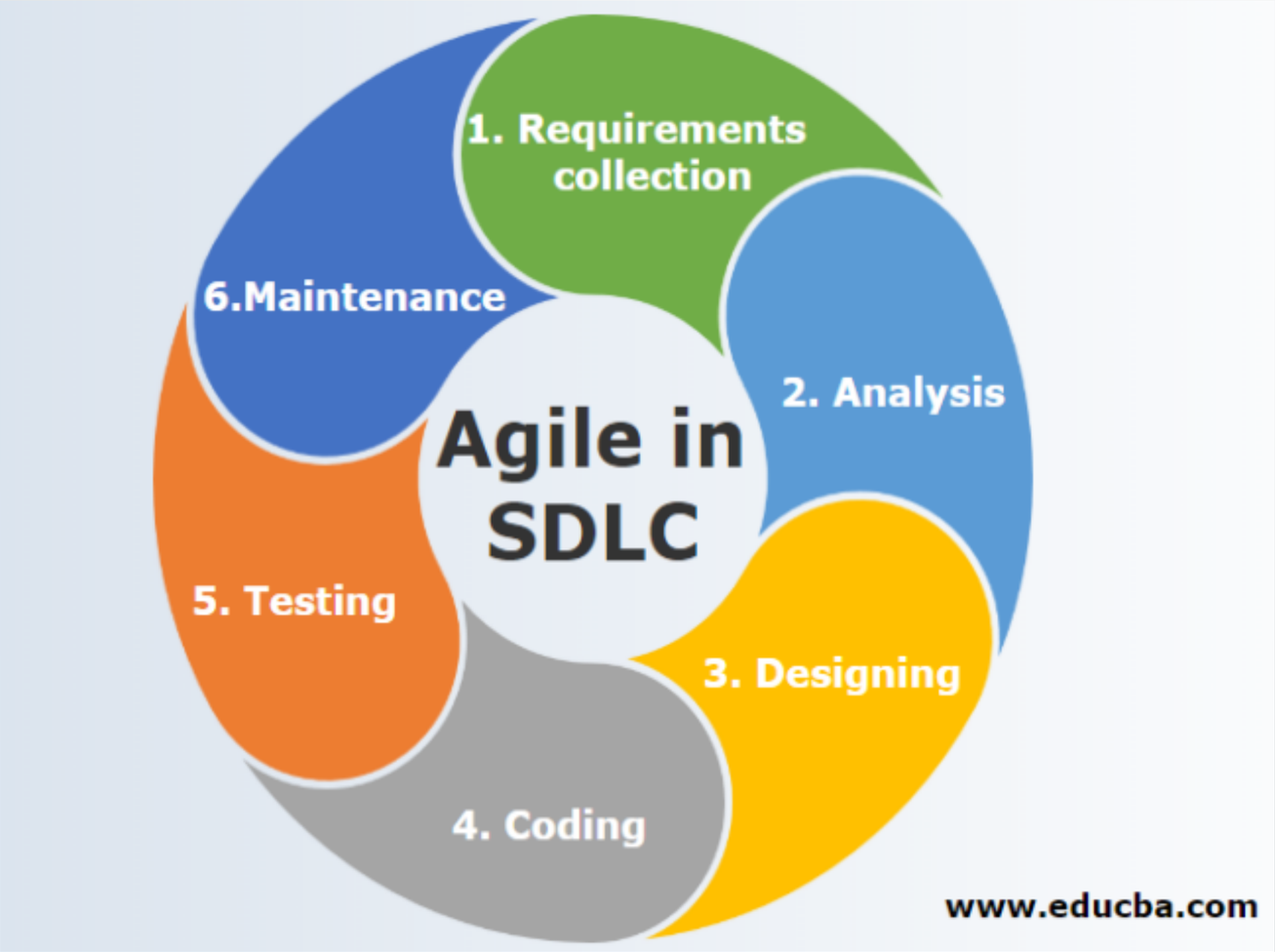
# HISTORY OF AGILE



# AGILE SOFTWARE DEVELOPMENT

Agile software development is an iterative and flexible approach to software development that prioritizes collaboration, customer satisfaction, and responding to change. It operates on the principles defined in the Agile Manifesto, which emerged in 2001 as an alternative to rigid, plan-driven methodologies like the Waterfall model.

# STEPS OF AGILE SDLC MODEL



The Agile model is a Combination of iterative and incremental Process models.
The Steps involve in agile SDLC models are;

1.**Requriments Collection**:Defining the initial project **vision**, goals, and high-level requirements. The team creates the **Product Backlog**– a prioritized, evolving list of desired features (often expressed as user stories).

2.**Analysis**:The team selects the highest-priority items from the Product Backlog to be completed in the upcoming sprint. They break down these features into smaller, manageable tasks and estimate the effort required .

3. **Designing**:Creating the design for the specific features to be built in the current sprint. This includes low-level design details like wireframes, database changes, and technical architectural adjustments needed for the increment.

**4.Coding**: Developers write and integrate the code for the selected features. This is done following continuous integration practices, where code is merged and built frequently.

**5.Testing**: Quality Assurance (QA) is performed on the newly developed increment. Testing is continuous and integrated into the sprint itself to ensure the new features are functional, bug-free, and meet the defined acceptance criteria.

**6.Maintenance**: The tested, working software increment is deployed to the end-users (or a staging environment). **A review/retrospective** meeting is held to gather feedback and reflect on the process, which then feeds back into the **Requirements Collection** for the next sprint.

## Some Key Principles and practices associated with agile methods in Software engineering:

NO GLYPH NO GLYPH NO GLYPH NO GLYPH NO GLYPH NO GLYPH NO GLYPH NO GLYPH NO GLYPH NO GLYPH NO GLYPH NO GLYPH

- Iterative and Incremental approach.
- Customer Centric approach.
- Collaborative Teams.
- Adoptive to Changes.
- Frequent deliveries.
- Continuous Testing and Quality Assurance.
- Prioritization.
- Backlog Management.
- Daily Standup meeting.
- Retrospectives(demonstration)

# Agile method Examples/types of agile frameworks

- Scrum
- Kanban
- Crystal
- Dynamic Software Development Methods(DSDM)
- Lean Software Development
- eXtreme Programming(XP)

NO GLYPH NO GLYPH NO GLYPH NO GLYPH

**Scrum:** Scrum is a lightweight framework for developing and Sustaining Complex Products through Collaboration, Accountability, and iterative Progress.

**Kanban:** Kanban is a Visual System for managing and improving the flow of Work. Its principles are drawn from lean manufacturing and focus on continuous delivery.

**Crystal:** The Crystal family, developed by Alistair Cockburn, consists of several methodologies (such as Clear, Yellow, Orange) that are tailored to the team Size and Project critically.

**eXtreme Programming(XP):** XP is an Agile framework focused on improving software quality and responsiveness to changing customer requirements. It is known for taking traditional development practices to "extreme" levels.

**Dynamic Software Development Method(DSDM):** The most prominent "dynamic" software development method is the **Dynamic Systems Development Method (DSDM)**. It is a comprehensive Agile framework that predates the Agile Manifesto and was created to provide a more disciplined approach to the iterative and rapid prototyping of the Rapid Application Development (RAD) method. DSDM is particularly known for its strong focus on project management, and it covers the full project lifecycle.

**Lean Software Development(LSD):** Derived from lean manufacturing ,LSD focuses on delivering maximum Customer value with minimum Waste. Its Seven principles emphasize efficiency, quality, and respect for people.

# Advantages of agile software development

- **Customer-Centric Focus: Highest priority is customer satisfaction** through early and continuous delivery of valuable working software.
- **Adaptability & Flexibility: Welcomes changing requirements,** even late in development, allowing the product to pivot to meet market needs.
- **Early Value Delivery:** Delivers **working software frequently** (in weeks rather than months), providing early return on investment (ROI).
- **Reduced Risk: Continuous integration and testing** in short cycles identify and resolve issues early, minimizing large-scale failures.
- **Improved Quality:** Focus on **continuous attention to technical excellence** and good design throughout every iteration.
- **Enhanced Team Collaboration:** Promotes **close, daily cooperation** between business and development teams, improving communication and morale.

# Disadvantages of agile software development

- **Less Predictability**: Difficulty in accurately estimating the **final project cost and timeline** upfront due to evolving requirements.

- **High Customer Dependency**: Requires **significant and constant time commitment** from the Product Owner or customer representative for feedback.

- **Documentation Deficit**: Emphasis on working software can lead to **minimal or incomplete documentation**, posing challenges for maintenance and new team members.

- **Risk of Scope Creep**: The flexibility to accept changes can lead to **uncontrolled expansion of the project's scope** if not managed with strict prioritization.

- **Requires Experienced Team**: Demands **high-level discipline, decision-making skills, and collaboration** from the development team and stakeholders to succeed

# When to use the agile model?

- The Agile Model is the ideal choice for projects with the following characteristics:
- **Evolving Requirements** 🔄: When user needs or market demands are likely to **change over the course of the project**. Agile processes welcome changes even late in development.
- **High Uncertainty/Innovation** 💡: For projects breaking new ground or requiring **creative problem-solving**, where flexibility to pivot and test solutions quickly is essential.
- **Need for Early Delivery** 🚀: When there is a business need to **release a functional product or valuable features quickly and frequently** (weeks instead of months) to market to achieve an early return on investment (ROI).
- **Customer Collaboration** 🗣️ : When the customer or end-user is willing and able to be **actively involved and provide continuous feedback** throughout the development lifecycle.
- **Smaller, Cross-Functional Teams** 🧑‍💻: When the project can be managed by **self-organizing, highly communicative teams** that can execute planning, design, coding, and testing within each short iteration.

# Thank you

*Adapt it with your needs and it will capture all the audience attention.*